

TAUFIQUL HAQUE KHAN TUSAR

GRADUATE STUDENT · AI & BACKEND ENGINEER

Dhaka 1750, Bangladesh

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Research Interest

My research interests span large language models, natural language processing, edge AI, and responsible ML. Specifically, I work on: (a) AI/ML applications in business, healthcare, and other complex domains; (b) security and privacy of AI systems for confidential data; (c) deployment in resource-constrained settings such as IoT and consumer devices.

Education

Bachelor of Science (B.Sc.) in Computer Science & Engineering

Apr 2018 – Nov 2022

City University

Dhaka 1216, Bangladesh

- **Thesis:** “Automated Detection of Acute Lymphoblastic Leukemia Subtypes from Microscopic Blood Smear Images Using Deep Neural Networks,” doi: [10.48550/arXiv.2208.08992](https://doi.org/10.48550/arXiv.2208.08992)
- **Research Focus:** Natural Language Processing, Healthcare AI, and Accessible AI.
- **Research Achievements:** 4 peer-reviewed publications (1 journal, 3 conferences).
- **CGPA:** 3.21 / 4.00

Research & Publications

Journal: 1 • Conference: 3 • Pre-print: 1 • Citations: 150+ • h-index: 5

[1] **M. T. H. Khan Tusar** et al., “An Intelligent Telediagnosis of Acute Lymphoblastic Leukemia using Histopathological Deep Learning,” Journal of Computing Theories and Applications (JCTA), vol. 2(1), pp. 1–12, 2024. doi: [10.62411/jcta.10358](https://doi.org/10.62411/jcta.10358).

[2] J. Hossain, M. T. Islam, and **M. T. H. K. Tusar** “Streamlining Brain Tumor Classification with Custom Transfer Learning in MRI Images,” 2023 IEEE Smart Information Systems and Technologies (IEEE SIST), Astana, Kazakhstan, 2023, pp. 522–526, doi: [10.1109/SIST58284.2023.10223507](https://doi.org/10.1109/SIST58284.2023.10223507).

[3] **M. T. H. Khan Tusar**, M. T. Islam, and A. H. Sakil, “An experimental study for early diagnosing Parkinson’s disease using machine learning,” Pre-print, 2023. doi: [10.48550/arXiv.2310.13654](https://doi.org/10.48550/arXiv.2310.13654).

[4] **M. T. Haque Khan Tusar**, M. T. Islam and F. I. Raju, “Detecting Chronic Kidney Disease (CKD) at the Initial Stage: A Novel Hybrid Feature-selection Method and Robust Data Preparation Pipeline for Different ML Techniques,” 2022 5th IEEE International Conference on Computing and Informatics (ICCI), New Cairo, Cairo, Egypt, 2022, pp. 400–407, doi: [10.1109/ICCI54321.2022.9756094](https://doi.org/10.1109/ICCI54321.2022.9756094).

[5] **M. T. H. K. Tusar** and M. T. Islam, “A Comparative Study of Sentiment Analysis Using NLP and Different Machine Learning Techniques on US Airline Twitter Data,” 2021 IEEE International Conference on Electronics, Communications and Information Technology (ICECIT), Khulna, Bangladesh, 2021, pp. 1–4, doi: [10.1109/ICECIT54077.2021.9641336](https://doi.org/10.1109/ICECIT54077.2021.9641336).

Research Experience

Research Engineer

Feb 2023 - Mar 2023

AIMS Lab, United International University

Dhaka 1212, Bangladesh

- Developed the backend for a multi-modal AI receptionist integrating face, voice, and speaker recognition.
- Implemented the referenced research work (doi.org/10.1109/ETCCE54784.2021.9689831) into a real-world application.

Undergrad Research Assistant

Dhaka 1216, Bangladesh

City University (Supervisor: Md. Touhidul Islam)

Dec 2020 - Nov 2022

- Contributed to machine learning research in healthcare (leukemia, Parkinson’s, CKD) and NLP applications.
- Trained and fine-tuned deep learning models; co-authored four peer-reviewed publications.
- Deployed two machine learning models into serviceable Flask applications.

Industry Experience

AI and Application Engineer

KafKai Giken Inc.

May 2023 - present

Tokyo 110-0016, Japan (Remote)

- Built and maintained the Django-PostgreSQL service and AI agents of [KAFKAI V3](#).
- Reduced LLM API costs by 50% by offloading/routing to task-specific capable models and providers.
- Built the FastMCP server, implemented action-based credit usage, and Stripe top-up for the upcoming Kafkai V4.
- Fine-tuned and maintained in-house LLMs from HuggingFace for long-running tasks, API cost-reduction, and data safety.
- Maintained the legacy Pyramid-based codebase of Kafkai V2 and converted it to Django-PostgreSQL architecture.
- Managed the deployment, monitoring, sentry, kanban, and technical support of multiple SaaS products.
- Collaborated with UI/UX team to improve user experience of [Kafkai.com](#), [Kafkai.ai](#), and [Puchiden.app](#).
- Conducted product cost and revenue analysis, pricing strategies, and competitor studies for [Kafkai.com](#) and [KaiMail.net](#).
- Managed the affiliate and partner programs of [Puchiden.app](#).
- Conducted technical interviews and the hiring process for the Engineering team.

Machine Learning Engineer (Intern)

Shadhin Lab LLC

July 2022 - Sept 2022

Dhaka 1219, Bangladesh

- Developed NLP-based large volume data collection, labeling, and classification pipelines for an in-house product.

Additional Experience

Independent Product Engineer

ChatCopilot.app

Feb 2026 - Present

Gazipur 1750, Bangladesh

- Architected and built ChatCopilot.app, a multi-tenant Django SaaS for AI-assisted sales, customer support, product discovery, orders, and appointment bookings.
- Implemented an agent runtime with tool calling, conversation state, history compaction, model routing, Bengali translation, image understanding, audio transcription, and human handover.
- Built tenant-scoped RAG for business knowledge and product/service catalogs using pgvector and multimodal embeddings.
- Designed integrations for Meta Graph API, website chat widget, Google Calendar, custom business ERP calendars, outbound webhooks, and Pathao courier dispatch.
- Implemented subscription plans, per-tenant quotas, LLM usage and cost accounting, sandbox testing, asynchronous message batching, and operational dashboards.
- Managed the DigitalOcean production droplet, production monitoring, resource usage, and PostgreSQL database.

Teaching Experience

Assistant Teacher

Milton English & Math Academy

Gazipur 1750, Bangladesh

Nov 2019 - Mar 2020

- Taught Physics and Information & Communication Technology to secondary and higher-secondary students.
- Mentored students in fundamental computational and problem-solving concepts.

Conferences

- **Reviewer**, 3rd International Conference on Computing and Information Technology (ICIT), 2023, Tabuk, Saudi Arabia
- **Technical Committee Member**, International Conference on Electronics, Information and Industrial Engineering, 2024, Wuhan, China.
- **Presenter**, 5th International Conference on Computing and Informatics (ICCI), 2022, Cairo, Egypt
- **Presenter**, International Conference on Electronics, Communications and Information Technology, 2021, Khulna, Bangladesh

Achievements

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|---|-------------|
| • Best Paper Award , 2023 IEEE International Conference on Smart Information System and Technologies | 2023 |
| • Merit Scholarship , City University, Bangladesh | Summer 2020 |
| • Junior School Government Scholarship , Ministry of Education, Bangladesh | 2012 |
| • Primary School Government Scholarship , Ministry of Primary and Mass Education, Bangladesh | 2009 |

Technical Skills

Languages	Python, SQL, Bash, C
GenAI	Transformers, PyTorch, LangGraph, FastMCP, llama-server, pgvector, Chroma
LLM APIs	OpenAI, Anthropic, OpenRouter, Perplexity, JinaAI, Ollama
ML & Data	scikit-learn, NLTK, OpenCV, pandas, NumPy, SciPy, Matplotlib
Backend	Django, FastAPI, Flask, Wagtail, Celery, Redis, Pytest, PostgreSQL, SQLite, Stripe
Deployment	AWS (EC2, Lightsail, S3), DigitalOcean, Render, Cloudflare (R2, Workers, D1, etc.), Caddy, Nginx
Others	REST APIs, OAuth, Webhooks, Git & GitHub, Linux, LaTeX, HTML, HTMX, Ajax, Jira/ClickUp

Technical Projects

Python Buddy 🧙

- Fine-tuned a small language model (Qwen2.5-coder-1.5-instruct) for code assistance, worked on low-resource training and inference.
- Achieved 7–12 tokens/s inference speed with 8-bit quantization on Snapdragon 685.
- **Tools:** HuggingFace Transformers, Llama.cpp, Python

Inference API Server 🧙

- Developed a lightweight Python API server for local/self-hosted LLM inference.
- Supports GGUF models with both string-based and role-based prompts, returns OpenAI-compatible responses.
- **Tools:** Python, HuggingFace Hub, Llama-cpp-Python

Bengali AI Receptionist 🧙

- Developed an AI receptionist Flask app integrating face recognition, speech-to-text, and text-to-speech.
- Verifies registered users and answers user queries using semantic similarity search.
- **Tools:** OpenCV, Sentence-Transformers, Flask, gTTS

Blood Cancer Stage Detection App 🧙

- Built a deep learning web app for acute lymphoblastic leukemia (ALL) stage detection.
- Finetuned and evaluated 4 CNN architectures; MobileNetV2 achieved 98% accuracy across 3 leukemia subtypes.
- **Tools:** TensorFlow, Flask, OpenCV, Unicorn, Render

Chronic Kidney Disease Diagnosis App 🧙

- Built a Flask-based ML app for early chronic kidney disease detection.
- Used LOF for outlier removal, SMOTE for imbalance handling, and Random Forest achieved maximum accuracy on test data.
- **Tools:** Scikit-learn, Flask, Unicorn, Joblib

Professional Memberships

• Student Member , Institute of Electrical and Electronics Engineers (IEEE)	08/2021 - 12/2022
• Member , IEEE Computer Society	08/2022 - 12/2022
• Executive Member , City University Computer Club	12/2020 - 10/2022
• Chapter Influencer , Google Crowdsourcing	09/2021 - 06/2022

Community Involvement

Community Lead

AI/ML Professional Community Bangladesh

March 2025 - Present

Bangladesh

- Organize technical webinars and workshops on AI/ML applications.
- Mentor early-career practitioners in development and career pathways.

Community Organizer

Python Bangladesh

Jan 2025 - Present

Bangladesh

- Organize tech meetups and collaborative events for Python developers across Bangladesh.
- Lead initiatives to restart PyCon Bangladesh.

Artificial Intelligence Mentor

The Empty Brain

June 2022 - March 2023

Dhaka 1212, Bangladesh

- Led technical teams in a startup incubator, ranked at top 25 in StartKoro Season 3.
- Mentored 20+ undergraduates in python, machine learning, and research methodologies.

JR. Machine Learning Engineer

Omdena

Dec 2022 - Feb 2023

CA 94301, USA (Remote)

- Enhanced a real-time Smart Traffic system with computer vision to reduce congestion on urban motorways.
- Collaborated with a global team (50+ members) to collect and process traffic image data.

Data Science Mentor

AI Planet (Formerly DPHI)

Sept 2021 - Oct 2021

Hyderabad, India (Remote)

- Supported the core team in running the data science boot-camp.
- Guided participants on Exploratory Data Analysis, created and evaluated final assessments.

Tech Events

- **Speaker**, PyCon ID 2025, Jakarta, Indonesia (14-15 December 2025)
- **Reviewer**, EuroPython 2025, Prague, Czechia (14-20 July 2025)

Extracurricular Activities

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|---|-------------------|
| • V-Force Volunteer , United Nations Volunteer | 08/2021 - 09/2021 |
| • Cadet , Bangladesh National Cadet Corps (BNCC) | 08/2015 - 01/2017 |
| • Scout , Bangladesh Scouts | 01/2008 - 12/2009 |

References

Kamal Mustafa

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Former Lecturer, Dept. of CSE, City University, Dhaka 1216, Bangladesh.

Sr. Software Engineer, Samsung R&D Institute, Dhaka, Bangladesh.

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